Generalized Linear Models. Count data

Modelització Estadística

Grau en Intel·ligència Artificial

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Introduction

- This lesson addresses the issue of count data via different models.

Example

Processes directly derived from counts:

- **Number of daily theft reports** in a given region.
- **Number of scientific papers** published by PhD students.
- **Number of damages** to a fleet of ships over the years.

Example

Processes derived from continuous variables (try to avoid):

- **Weekly study hours** by students. It could be considered continuous if measured with enough precision, and a linear model could be applied.
- **Creditworthiness**. Credit companies quantify it ordinally. Preferably, an ordinal logistic model should be applied.

Poisson Model. Properties

- It is known as a **log-linear** because it is usually formulated via the canonical **logarithmic link**.

$$log(\mu_i) = x_i^T \beta \qquad i = 1, \ldots, n$$

- The dependence of the expected value of the response (μ_i) conditioned on a vector of covariates and/or factors (x_i) is **multiplicative** (i.e, an additive increment in the explanatory variable has a multiplicative influence on the response)

Key property

It is assumed that the **conditional variance** of the response is **equal to the expected value**:

$$V(Y_i|X_i) = \mu_i$$

Review of Distributions. Poisson

- **Definition**: Number of events in a given time and/or space interval
- **Notation**: $X \sim P(\lambda)$
- **Parameters**: λ (rate of occurrence) with $\lambda > 0$

- **Probability function**:

  $P(X = K) = \frac{e^{-\lambda} \cdot \lambda^k}{k!}$ with $k = 0, 1, \cdots$
- **Indicators**:
  - $E(X) = \lambda$
  - $V(X) = \lambda$
- **R**: *dpois, ppois, qpois*

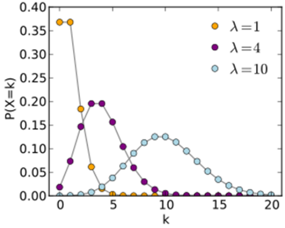


Figure 1: Poisson distribution

Review of Distributions. Negative Binomial

- **Definition**: Number of k repetitions of a Bernoulli experiment until r successes are obtained
- **Notation**: $X \sim NB(r, p)$
- **Parameters**: r (number of successes), p (probability of success)

- **Probability function**:

$$P(X = K) = \binom{k-1}{r-1} \cdot p^r \cdot q^{k-r}$$
$$k = r, \cdots, n$$

- **Indicators**:
  - $E(X) = \frac{r \cdot q}{p}$
  - $V(X) = \frac{r \cdot q}{p^2}$
- **R**: *dnbinom, pnbinom, qnbinom*

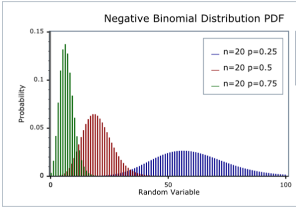


Figure 2: Negative Binomial distribution

Poisson. Example (I)

- Number of defects in a routine inspection of houses.

Defects	Houses (Obs.)	%	Poisson Prob.	Houses (Exp.)	Chi-square
0	18	0.041	0.047	20.89	0.4
1	53	0.120	0.145	63.65	1.78
2	103	0.234	0.220	97.00	0.37
3	107	0.243	0.224	98.54	0.73
4	82	0.186	0.171	75.08	0.64
5	46	0.105	0.104	45.77	0
6	18	0.041	0.053	23.25	1.18
7	10	0.023	0.023	10.12	0
8	2	0.005	0.009	3.86	0.89
9	1	0.002	0.003	1.31	0.07

Poisson. Example (II)

- Estimated expected mean ($\hat{\lambda} = \hat{\mu}$)

$$\hat{\lambda} = \frac{\textit{no. defects}}{\textit{no. houses}} = \frac{0 \cdot 18 + 1 \cdot 53 + \cdots + 9 \cdot 1}{18 + \cdots + 1} = \frac{1341}{440} = 3.05$$

- The Pearson statistic **compares** the observed defects with those expected according to a Poisson distribution.
- The fit is good enough since the p-value is **higher** than 0.05:

$$\chi^2 = 0.4 + \cdots + 0.07 = 6.07 \ ; \ df = 10 - 1 - 1 = 8 \rightarrow p \ value = 0.63$$

Explanation

- **Pearson χ^2 statistic** assesses how well the Poisson model fits the data.
- This statistic compares the **observed** frequencies of defects with the frequencies **expected** under the Poisson model.
- The statistic is the **sum of the chi-square values for each category**
- The statistic follows a χ^2 **distribution** under H_0 with degrees of freedom equal to the number of categories minus the number of parameters minus 1.
- A **p-value** $<<$ 0.05 would suggest that the model doesn't fit the data well.

Poisson Model. Assumptions

The following assumptions should be met to fit a Poisson model.

1. The Poisson distribution is **discrete** with a single parameter (λ or μ). The parameter represents the expected number of events per unit of time.
2. The response (y) must be **non-negative integers**; that is, $Y \geq 0$.
3. The observations are **independent** one of each other.
4. The **mean and variance** of the model should be **similar**.
5. The **Pearson dispersion statistic** χ^2 should have a **value close to 1**. A value of 1 is obtained when the observed and predicted variances of the response are the same.

Poisson Model. Numerical Validation

Pearson Statistic

$$X^2 = \sum_{i=1}^{n} \frac{(y_i - \hat{\mu}_i)^2}{\hat{\mu}_i} \sim \chi^2_{J-p-1}$$

Deviance Statistic

$$D = 2\sum_{i=1}^{n} \left\{ y_i \log\left(\frac{y_i}{\hat{\mu}_i}\right) \right\} \sim \chi^2_{J-p-1}$$

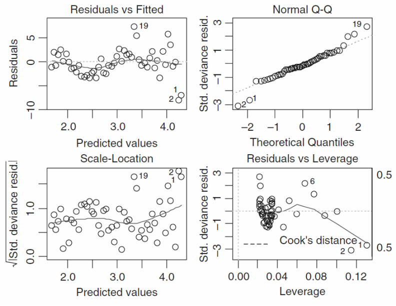
[**J**: number of intervals used to calculate the statistic; **p**: number of parameters]

- Both tests are based on comparing our model with the **saturated model**.
- Both tests are **asymptotically equivalent** and should reasonably align for large sample sizes.
- If the expected **values per cell are less than 5**, it is preferable to use the deviance statistic rather than Pearson's statistic.

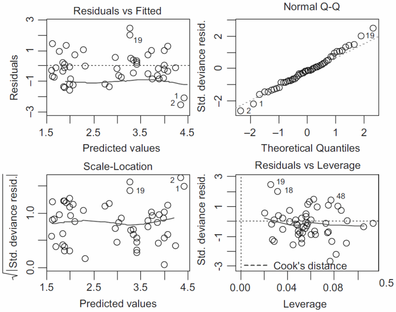
Poisson Model. Graphical Validation

- Types of residual plots (no patterns should be observed):
  - Residuals vs. predicted values. Heteroscedasticity. **Is there overdispersion?**
  - Residuals vs. explanatory variables. **Should they be transformed?**
  - Residuals vs. variables not included in the model. **Should they be included?**
  - Residuals vs. time. **Does the independence assumption hold?**
- The goals of residual analysis are:
  - Discover patterns. **E.g., quadratic terms**
  - Identify outliers. **E.g., very large Cook's distances**
  - Identify "errors" in the model specification. **E.g., overdispersion**

Non-validated model



Validated model



Poisson Model. Interpretation of cofficients

Intercept term

- In the null model, the exponential of the intercept is the estimated **global mean**.
- In a non-null model, the exponential of the intercept is the **estimated mean in the group with all reference categories and zero values in the covariates**.

Coefficient of a factor

The exponential of the coefficient is the **multiplicative increase (or decrease) in the estimated mean of the response** in the relevant category compared to the baseline category. It is a **rate ratio**.

Coefficient of a covariate

The exponential of the coefficient is the **multiplicative increase (or decrease) in the estimated mean of the response** for each unit increase in the covariate. It is also a **rate ratio**.

Poisson Model. Offset (I)

- Sometimes, it is more appropriate to model **rates** instead of raw counts.

Example

- When the **follow-up time** for counting events is different across observations. For example, *the number of psoriasis outbreaks in a set of patients with different follow-up periods*.
- When dealing with **aggregated data** where groups are unbalanced. For example, *the number of Legionella cases across different regions*.

Offset

In abovementioned cases, the **offset** is introduced into the model, usually in logarithmic form, to account for the differing exposure times (e.g., follow-up time or population size). This allows us to model the rate (λ), rather than just the raw counts.

$$log(y_i) = log(t_i) + x_i^T \beta \rightarrow log\left(\frac{y_i}{t_i}\right) = log(\lambda_i) = x_i^T \beta$$

Poisson Model. Offset (II)

Offset

- The **offset** is essentially a correction factor that adjusts for varying exposure times (t_i) or group sizes (n_i). In a Poisson model, we assume that the number of events is proportional to the length of time or size of the population being studied.
- By including the offset, we standardize the outcome to a **rate per unit** (e.g., per person, per day, or per region), which provides a clearer interpretation. The model then estimates the effect of the covariates on the rate of events, rather than the raw count.
- For instance, if patients have different follow-up times, directly modeling the counts without an offset could bias the results, as longer follow-up periods would naturally lead to more events. The offset ensures that the model estimates the true **rate of occurrence**.

Poisson Model. R

- To model a **Poisson-distributed response**, use the glm function with the parameter family=poisson

```
mod=glm(response ~ pred1 + ... + predk, family=poisson)
mod=glm(response ~ pred1 + ... + predk, offset=log(time),  family=poisson)
mod=glm(response ~ offset(log(time))+ pred1 + ... + predk, family=poisson)
```

- The summary function provides **more information** about the model:

```
summary(mod)
```

- The deviance and residuals functions allow us to obtain the **Deviance** and **Pearson residuals**, respectively:

```
deviance(mod)
residuals(mod,"pearson")
```

Poisson Model. Overdispersion Problem

- The **assumption** of the Poisson model that the conditional variance and mean are equal ($\phi = 1$) may not always hold.

$$V(Y_i|X_i) = \phi \cdot \mu_i$$

- The **dispersion parameter** can be greater (common) or less (uncommon) than 1.

Underdispersion	Equidispersion	Overdispersion
$\phi < 1$	$\phi = 1$	$\phi > 1$

- Poisson models are infrequent because there are usually **additional sources of variability** that cause overdispersion.

Example

- **Variable follow-up time**. It could be addressed using an offset.
- Data from a **clustered process** with different characteristics.

Poisson Model. Overdispersion. Estimation

- One way to estimate the **(over)dispersion parameter** is by using the Pearson statistic divided by the degrees of freedom:

$$\hat{\phi} = \frac{X^2}{n-p} = \frac{\sum_{i=1}^{n} \frac{(y_i - \hat{\mu}_i)^2}{\hat{\mu}_i}}{n-p}$$

$\hat{\phi} >> 1$ indicates overdispersion.

What does a high value of the Pearson statistic indicate?

It might indicate two different things:

- **Apparent overdispersion**: It is a **model misfit** because of the **absence** of important covariates and/or interactions, the **presence** of outliers or unaccounted nonlinear effects, or an incorrect choice of the link function.
- **True overdispersion**: The variability is greater than the mean due to an excess number of zeros, clustered or correlated observations.

Poisson Model. Overdispersion. Solutions

We can use different models in case of overdispersion in the Poisson model.

- **Quasi-Poisson model**. The coefficients are estimated using the Poisson distribution, but the standard errors are adjusted to account for $\phi \neq 1$.

- **Negative binomial model**. The variance depends quadratically on the mean:

$$V(Y_i|X_i) = \mu_i + \frac{1}{\theta}\mu_i^2$$

- **Zero-inflated model**. If the issue is an excess of zeros, we can consider two processes at once: one that generates the zeros and another that generates the counts.

- **Zero-truncated model**. When zeros are not part of the observed data but theoretically possible.

Quasi-Poisson

We can estimate a quasi-poisson model by following the next steps:

1. An initial estimate of the **(over)dispersion parameter** ϕ is calculated:

$$\hat{\phi} = \frac{\chi_P^2}{n-p} \text{ where } \chi_P^2 \text{ is the Pearson statistic.}$$

2. The **point estimates** of the β coefficients remain the same as in the Poisson model.

3. The **standard errors** are estimated using this dispersion parameter. The standard errors of the Poisson model are uniformly inflated by $\sqrt{\hat{\phi}}$.

- In R, the function `glm` is used with the `family=quasipoisson` parameter:

```
mod=glm(response ~ pred1 + ..., family = quasipoisson)
mod=glm(response ~ pred1 + ..., family = quasi(link="log")) # Equivalent
```

Negative Binomial. Introduction (I)

- Another approach to handle overdispersion is to start with a Poisson distribution, assuming there is a multiplicative random effect Z that represents **unobservable** heterogeneity.

- The distribution of the response, conditional on a set of unobserved variables W, follows a **Poisson distribution with rate μZ**:

$$Y|W \sim P(\mu Z)$$

- We must make an **assumption** about Z:

$$Z \sim Gamma\left(\alpha = \frac{1}{\sigma^2}, \beta = \frac{1}{\sigma^2}\right) \rightarrow E(Z) = \frac{\alpha}{\beta} = 1 \text{ and } V(Z) = \frac{\alpha}{\beta^2} = \sigma^2$$

- It implies:

$$V(Y_i) = \mu_i \cdot (1 + \sigma^2 \mu_i) = \mu_i + \frac{\mu_i^2}{\theta}$$

- The parameter $\theta = 1/\sigma^2$ can be estimated:
  - High values indicate little overdispersion
  - Low values indicate overdispersion

Negative Binomial. R

- The first step is to estimate **the parameter θ** using the function `glm.nb` from the `MASS` package:

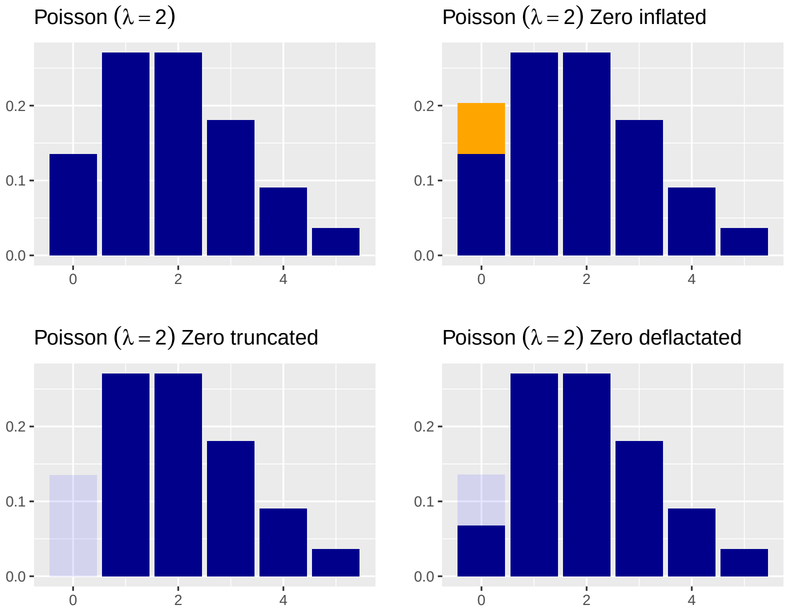
```
mod1  <- glm.nb(response ~ pred1 + ... + predk)
Theta <- mod1$theta
```

- This estimation of θ is then used to **formulate the full model** with `glm` and the `family = negative.binomial(Theta)` parameter:

```
mod2 <- glm(response ~ pred1 + ... + predk, family=negative.binomial(Theta))
```

- The point estimates from both functions (*glm* and *glm.nb*) are the same.

Different problems with zero counts



Zero-Inflated Models. Introduction

- A common issue in count models arises when empirical data shows **more zeros** than expected under certain models (e.g., Poisson and Negative Binomial).

Example

- **Number of murders** committed by a person in a year.
- **Number of military conflicts** a country has been involved in during a year.

- In general, this occurs with **events that are rare for some observations** and **relatively common for others**.
- It is **hard to predict in advance** whether this kind of models will be necessary.

Zero-Inflated Models. Introduction

Base

Let $f(y)$ be the count density function. To formulate a zero-inflated model, a separate component to inflate the probability of observing a zero is added:

$$P(Y_i = k) = \begin{cases} \pi + (1-\pi)\cdot f(0) & \text{if } k = 0 \\ (1-\pi)\cdot f(k) & \text{if } k > 0 \end{cases}$$

Example

The **Zero-Inflated Poisson** (ZIP) model includes a proportion π of ***structural zeros*** and a proportion $(1-\pi)$ of the population that follows a Poisson distribution:

- $P(Y_i = 0) = \pi + (1-\pi)\cdot e^{-\lambda}$
- $P(Y_i = k) = (1-\pi)\cdot \frac{\lambda^k e^{-\lambda}}{k!}$ for $k \geq 1$

- The proportion of zeros π is added to the baseline distribution (Poisson or Negative Binomial), and the probabilities of the base model are reduced by the proportion $(1-\pi)$ to ensure that the probabilities sum to one.

Zero-Inflated Models. Models

- In some cases, a possible cause of excess zeros is **misrecorded observations**, such as values not detected by a machine.

- The probability π can either be set as a **constant** or modeled using a **binary response model** to predict it.

Modelization

In general, two models are combined:

- A **logit model** that predicts the probability of belonging to one of the two latent classes for each observation.
- A **Poisson** or **Negative Binomial model** that predicts the response for those individuals in the second latent class.

- Ignoring the excessive presence of zeros can lead to biased results and/or overdispersion.

Zero-Inflated Models. Types of zeros

- In these models, there are two types of zeros: **structural** and **random**.

Example

In a simple random sample of Barcelona residents, one might study the **number of times they have won the lottery** over a given period. **Structural zeros** would come from the portion of the population that never plays the lottery. Among those who have played at least once, **random** zeros can also occur if they have never won.

Example

The **number of cigarettes smoked** over a period of time would include **structural** zeros for non-smokers.

Zero-Inflated Models. R (I)

- The function `zinfl` from the `pscl` package is an option to estimate a zero-inflated Poisson model:

```
mod.poisson.zinfl <- zinfl(response ~ pred1 + ... + predk)
summary(mod.poisson.zinfl)
```

- The same function can be used to fit a zero-inflated negative binomial model:

```
mod.nb.zinfl <- zinfl(response ~ pred1 + ... + predk, dist = 'negbin')
summary(mod.nb.zinfl)
```

Zero-Inflated Models. R (II)

- The `zeroinfl` function from the `countreg` package (*Github*) is another option to estimate a zero-inflated Poisson model:

```
mod.poisson.zinfl <- zeroinfl(response ~ pred1 + ... + predk,
                              dist = "poisson", link = "logit")
summary(mod.poisson.zinfl)
```

- The same function can be used to fit a zero-inflated negative binomial model:

```
mod.nb.zinfl <- zeroinfl(response ~ pred1 + ... + predk,
                         dist = "negbin", link = "logit")
summary(mod.nb.zinfl)
```

- It allows us to fit models with different variables for both parts (zeros and counts)

Zero-Truncated Models. Introduction

- **Truncation occurs** when the observations (y_i, x_i) are completely missing within a certain range of y_i.

- Truncation can be **left-truncation** (when values $y_i < k$ are not observed) or **right-truncation** (when values $y_i > k$ are not observed).

- The most common case of left-truncation involves **zero-truncated** data, where the response variable cannot take the value **0**.

Example

- Days of hospitalization for patients
- Number of dolphins in a group
- Weekly bus usage in a survey conducted on buses

- It can be **problematic** to fit Poisson or Negative Binomial models when there are no zeros, as these distributions allow zeros in their range. It could be produced biased results.

Zero-Truncated Models. Introduction

- The key is to modify the probability function to account for non-observed zeros.

Example

In a standard Poisson distribution:

$$P(X = K) = \frac{e^{-\lambda} \cdot \lambda^k}{k!} \text{ with } k = 0, 1, \cdots$$

In a zero-truncated Poisson distribution, the probabilities for strictly positive values are standardized:

$$P(X = 0) = e^{-\lambda} \rightarrow P(X = K | K > 0) = \frac{e^{-\lambda} \cdot \lambda^k}{(1 - e^{-\lambda}) k!}$$

- A similar process can be applied for the Negative Binomial distribution.
- Fitting a model without accounting for the absence of zeros leads to biased coefficient estimates and **underestimated standard errors**.

Zero-Truncated Models. R

- The VGAM package contains the vglm function to fit zero-truncated models.

```
library(VGAM)
vglm(formula, family = pospoisson, data)       # Zero-truncated Poisson
vglm(formula, family = posnegbinomial, data)  # Zero-truncated Negative Binor
```

- The countreg package (*Github*) is another option for fitting zero-truncated models.

```
# devtools::install_github("rforge/countreg/pkg")
library(countreg)
zerotrunc(formula, dist = "poisson")      # Zero-truncated Poisson
glm(formula, family = ztpoisson)          # Equivalent alternative
zerotrunc(formula, dist = "negbin")       # Zero-truncated Negative Binomial
```

Example. Scientific articles

- Variables
  - **art**: articles in last three years of PhD
  - **fem**: coded one for females
  - **mar**: coded one if married
  - **kid5**: number of children under age six
  - **phd**: prestige of Ph.D. program
  - **ment**: articles by mentor in last three years

- Sample (4 first rows)

art	fem	mar	kid5	phd	ment
0	Men	Married	0	2.52	7
0	Women	Single	0	2.05	6
0	Women	Single	0	3.75	6
0	Men	Married	1	1.18	3

Example. Scientific articles. Modelo Poisson (I)

```
mp <- glm(art ~ fem + mar + kid5 + phd + ment, family=poisson, data=ab)
summary(mp)
```

```
##
## Call:
## glm(formula = art ~ fem + mar + kid5 + phd + ment, family = poisson,
##     data = ab)
##
## Coefficients:
##               Estimate Std. Error z value Pr(>|z|)
## (Intercept)  0.304617   0.102981   2.958   0.0031 **
## femWomen    -0.224594   0.054613  -4.112 3.92e-05 ***
## marMarried   0.155243   0.061374   2.529   0.0114 *
## kid5        -0.184883   0.040127  -4.607 4.08e-06 ***
## phd          0.012823   0.026397   0.486   0.6271
## ment         0.025543   0.002006  12.733  < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for poisson family taken to be 1)
##
##     Null deviance: 1817.4  on 914  degrees of freedom
## Residual deviance: 1634.4  on 909  degrees of freedom
## AIC: 3314.1
##
## Number of Fisher Scoring iterations: 5
```

Example. Scientific articles. Modelo Poisson (II)

- Is this model valid? Does it fit the data well?

Example. Scientific articles. Modelo Poisson (II)

- Is this model valid? Does it fit the data well?

R

```
qchisq(0.95, df.residual(mp))
```

```
## [1] 980.2518
```

Answer

The model is not good enough since the residual deviation (**1634.4**) is clearly higher than the critical point ($\chi^2_{909} = 980.3$)

Example. Scientific articles. Overdispersion (I)

- Is there overdispersion?

Example. Scientific articles. Overdispersion (I)

- Is there overdispersion?

R

```
library(AER)
dispersiontest(mp,trafo=1)
```

```
##
##  Overdispersion test
##
## data:  mp
## z = 5.7825, p-value = 3.681e-09
## alternative hypothesis: true alpha is greater than 0
## sample estimates:
##     alpha
## 0.8245398
```

```
dispersiontest(mp,trafo=2)
```

```
##
##  Overdispersion test
##
## data:  mp
## z = 6.5297, p-value = 3.295e-11
## alternative hypothesis: true alpha is greater than 0
## sample estimates:
##     alpha
## 0.5091216
```

Answer

There is overdispersion that must be corrected.

Example. Scientific articles. Sobredispersión (II)

- Is there overdispersion?

Respuesta (cont.)

- Both the test to check whether the variance depends linearly on the expectation (*trafo=1*) and to see if it depends quadratically (*trafo=2*) are significant:
  - $trafo = 1 \rightarrow V[Y_i|X_i] = (1 + 0.82) \cdot E[Y_i|X_i]$
  - $trafo = 2 \rightarrow V[Y_i|X_i] = E[Y_i|X_i] + 0.51 \cdot E[Y_i|X_i]^2$

Example. Scientific articles. Quasi-Poisson

```r
summary(mq <- glm(art~fem+mar+kid5+phd+ment, family=quasipoisson, data=ab))
```

```
##
## Call:
## glm(formula = art ~ fem + mar + kid5 + phd + ment, family = quasipoisson,
##     data = ab)
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  0.304617   0.139273   2.187 0.028983 *
## femWomen    -0.224594   0.073860  -3.041 0.002427 **
## marMarried   0.155243   0.083003   1.870 0.061759 .
## kid5        -0.184883   0.054268  -3.407 0.000686 ***
## phd          0.012823   0.035700   0.359 0.719544
## ment         0.025543   0.002713   9.415  < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for quasipoisson family taken to be 1.829006)
##
##     Null deviance: 1817.4  on 914  degrees of freedom
## Residual deviance: 1634.4  on 909  degrees of freedom
## AIC: NA
##
## Number of Fisher Scoring iterations: 5
```

Example. Scientific articles. Quasi-Poisson

- What can you say about the output of the model?

Example. Scientific articles. Quasi-Poisson

- What can you say about the output of the model?

R

```
pr <- residuals(mp,"pearson")
phi <- sum(pr^2)/df.residual(mp)
round(c(phi,sqrt(phi)),4)
```

```
## [1] 1.8290 1.3524
```

Answer

- The AIC cannot be calculated because there is no complete probabilistic model.
- The estimate of ϕ by the Pearson statistic is almost identical to that provided by the `dispersion.test` (**1.82**).
- The quasi-poisson model uses the estimate by the Pearson statistic to correct the standard errors: the standard errors will be *inflated* by **1.35** $(= \sqrt{1.82})$.

Example. Scientific articles. Quasi-Poisson vs. Poisson

- What can you say about the output of the model?

R

```
se <- function(model) sqrt(diag(vcov(model)))
kable(round(data.frame(p=coef(mp),  q=coef(mq),
                       se.p=se(mp), se.q=se(mq),
                       ratio=se(mq)/se(mp)), 4))
```

	p	q	se.p	se.q	ratio
(Intercept)	0.3046	0.3046	0.1030	0.1393	1.3524
femWomen	-0.2246	-0.2246	0.0546	0.0739	1.3524
marMarried	0.1552	0.1552	0.0614	0.0830	1.3524
kid5	-0.1849	-0.1849	0.0401	0.0543	1.3524
phd	0.0128	0.0128	0.0264	0.0357	1.3524
ment	0.0255	0.0255	0.0020	0.0027	1.3524

Answer (cont.)

- The coefficients of the Poisson and Quasi-Poisson models are the same and the standard errors are multiplied by 1.35 which is obtained from the square root of the Pearson statistic divided by its degrees of freedom.

Example. Scientific articles. Negative Binomial

```r
summary(mnb <- glm.nb(art ~ fem + mar + kid5 + phd + ment, data=ab))
```

```
##
## Call:
## glm.nb(formula = art ~ fem + mar + kid5 + phd + ment, data = ab,
##     init.theta = 2.264387695, link = log)
##
## Coefficients:
##              Estimate Std. Error z value Pr(>|z|)
## (Intercept)  0.256144   0.137348   1.865 0.062191 .
## femWomen    -0.216418   0.072636  -2.979 0.002887 **
## marMarried   0.150489   0.082097   1.833 0.066791 .
## kid5        -0.176415   0.052813  -3.340 0.000837 ***
## phd          0.015271   0.035873   0.426 0.670326
## ment         0.029082   0.003214   9.048  < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for Negative Binomial(2.2644) family taken to be 1)
##
##     Null deviance: 1109.0  on 914  degrees of freedom
## Residual deviance: 1004.3  on 909  degrees of freedom
## AIC: 3135.9
##
## Number of Fisher Scoring iterations: 1
##
##
##               Theta:  2.264
##           Std. Err.:  0.271
##
##  2 x log-likelihood:  -3121.917
```

Example. Scientific articles. Negative Binomial

- Estimate of θ y aplication in `glm`

```
1/mnb$theta
```

```
## [1] 0.4416205
```

```
summary(mnbg <- glm(art ~ fem + mar + kid5 + phd + ment,family=negative.binomial(mnb$theta), data=ab))
```

```
##
## Call:
## glm(formula = art ~ fem + mar + kid5 + phd + ment, family = negative.binomial(mnb$theta),
##     data = ab)
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  0.256149   0.140008   1.830  0.06765 .
## femWomen    -0.216420   0.074043  -2.923  0.00355 **
## marMarried   0.150490   0.083686   1.798  0.07247 .
## kid5        -0.176416   0.053836  -3.277  0.00109 **
## phd          0.015271   0.036568   0.418  0.67633
## ment         0.029082   0.003277   8.876  < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for Negative Binomial(2.2644) family taken to be 1.03911)
##
##     Null deviance: 1109.0  on 914  degrees of freedom
## Residual deviance: 1004.3  on 909  degrees of freedom
## AIC: 3133.9
##
## Number of Fisher Scoring iterations: 6
```

Example. Scientific articles. Negative Binomial

R

```
deviance(mnbg)
```

```
## [1] 1004.281
```

```
qchisq(0.95,909)
```

```
## [1] 980.2518
```

Answer (cont.)

- The model does not fit the data well since the residual deviance is in the non-acceptance region of H_0
- The estimate of θ provided by the model (2.26) is similar to that provided by *dispersion.test* $(1.96 = 1/0.509)$.

Example. Scientific articles. Zero-inflated poisson

- In the zero-inflated Poisson model, we have 2 models in one:
  - A logit model (lower output of `zeroinfl`) to predict the probability of belonging to the structural zero class.
  - A Poisson model (upper output of `zeroinfl`) for the random zeros and the rest of the values.

Example. Scientific articles. Zero-inflated poisson. Exercise

```
summary(mzip <- zeroinfl(art ~ fem + mar + kid5 + phd + ment, data=ab))
```

```
##
## Call:
## zeroinfl(formula = art ~ fem + mar + kid5 + phd + ment, data = ab)
##
## Pearson residuals:
##     Min      1Q  Median      3Q     Max
## -2.3253 -0.8652 -0.2826  0.5404  7.2976
##
## Count model coefficients (poisson with log link):
##              Estimate Std. Error z value Pr(>|z|)
## (Intercept)  0.640838   0.121307   5.283 1.27e-07 ***
## femWomen    -0.209145   0.063405  -3.299 0.000972 ***
## marMarried   0.103751   0.071111   1.459 0.144565
## kid5        -0.143320   0.047429  -3.022 0.002513 **
## phd         -0.006166   0.031008  -0.199 0.842378
## ment         0.018098   0.002294   7.888 3.07e-15 ***
##
## Zero-inflation model coefficients (binomial with logit link):
##              Estimate Std. Error z value Pr(>|z|)
## (Intercept) -0.577059   0.509386  -1.133  0.25728
## femWomen     0.109746   0.280082   0.392  0.69518
## marMarried  -0.354014   0.317611  -1.115  0.26502
## kid5         0.217097   0.196482   1.105  0.26919
## phd          0.001274   0.145263   0.009  0.99300
## ment        -0.134114   0.045243  -2.964  0.00303 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Number of iterations in BFGS optimization: 21
## Log-likelihood: -1605 on 12 Df
```

1. What is the probability of structural zero for a single woman with no children under 6 years of age in a program with 3 prestige and 1 published article by her mentor?
2. What is the expected mean for this woman?
3. How much more likely is a woman to be a structural zero than a man?

Example. Scientific articles. Zero-inflated poisson. Solution

Answer 1

```
pr1 <- predict(mzip,newdata = data.frame(fem='Women',mar='Single',kid5=0,phd=3,ment=1),type='zero')     # With function
co <- coef(mzip)                                                                                         # By hand
pl <- co[7] + co[8] + 3*co[11] + 1*co[12]
pr2 <- exp(pl)/(1+exp(pl))
round(as.numeric(c(pr1,pr2)),3)
```

```
## [1] 0.355 0.355
```

Answer 2

```
mu1 <- predict(mzip,newdata = data.frame(fem='Women',mar='Single',kid5=0,phd=3,ment=1),type='response') # With function
pl0 <- co[1] + co[2] + 3*co[5] + 1*co[6]                                                                 # By hand
mu2_0 <- exp(pl0)
mu2 <- (1-pr2)*mu2_0
round(as.numeric(c(mu1,mu2)),3)
```

```
## [1] 0.993 0.993
```

Answer 3

```
round(as.numeric(exp(co[8])),3)    # Odds ratio
```

```
## [1] 1.116
```

References

1. Cameron, A.C., and Trivedi, P.K. (1998). Regression Analysis of Count Data. Cambridge: Cambridge University Press.
2. Zuur, AF, Ieno, EN, Walker NJ et al (2009). Mixed Effects Models and Extensions in Ecology with R. Springer.